

Unit 1 Quick Reference

Python syntax, one page. For the *why*, see the lesson reference sheets.
Covers Lessons 3–4.

Output

```
print("Hi")           # Hi
print("a", "b")       # a b           commas add a space
print("Sum", 7 + 3)   # Sum 10       math runs first

print("a", "b", sep="-") # a-b       what goes BETWEEN
print("a", "b", sep="")  # ab
print("Loading", end="") # no newline what goes AFTER
```

sep separates **every** pair, including after a label.

`print("R:", 1, 2, sep=" > ")` → `R: > 1 > 2`

Fix: `print(f"R: {1}", 2, sep=" > ")`

Repeating & widths

```
"-" * 40           # 40 dashes
"Hi! " * 3         # Hi! Hi! Hi!
len("PYTHON")      # 6
(40 - len(title)) // 2 # centering padding
```

`"*" * "20"` → **TypeError**. String × *number* only.

Quotes

```
print('She said "hi"')           # double inside single
print("It's fine")               # single inside double
print("He said \"it's mine\"")   # both -> escape
print("""He said "it's mine" """) # both -> triple quotes
```

f-strings

```
name, age = "Alex", 15

f"Hi {name}"           # Hi Alex
f"{age + 1}"           # 16           any expression
f"{name.upper()}"      # ALEX         method calls
f"{cost:.2f}"          # 29.29       2 decimal places
```

Forget the `f` and it prints the braces — **with no error**.

Escapes & multiline

```
"a\nb"           # newline
"a\tb"           # tab
"C:\\Users"      # one real backslash

'''line one
line two'''      # spans lines, no substitution
f'''{name} here
line two'''      # spans lines AND substitutes
```

	Spans lines?	Substitutes?
<code>"""..."""</code>	yes	no
<code>f"..."</code>	no	yes
<code>f"""..."""</code>	yes	yes

Variables

```
player = "Nova"           # no let, no const, no var
MAX_LEVEL = 99            # ALL CAPS = "don't change this"
student_name = "Alex"     # snake_case, never camelCase
```

Can't start with a number. No spaces. Case matters.

The four types

```
"Nova"    # str    text, in quotes
12         # int    whole number
82.5      # float  has a decimal point
True      # bool   True / False, capitalized, no quotes

type(player)           # <class 'str'>
type(42) vs type(42.0) # int vs float
```

Conversion

```
str(12)           # "12"
int("95")         # 95
float("3.85")     # 3.85

int(3.7)          # 3      CHOPS, never rounds
int("3.7")        # ValueError
int(float("3.7")) # 3      go through float
```

"5" + 3 → **TypeError**. JavaScript gives "53" ; Python refuses.
Fix: "5" + str(3) or f"5{3}"

Division

```
10 / 2           # 5.0    ALWAYS a float
10 // 2          # 5      whole number
10 // 3          # 3      chops remainder
```

Error → cause

Error	It means
<code>TypeError: can only concatenate str</code>	<code>+</code> between text and a number
<code>TypeError: can't multiply sequence</code>	<code>string × string</code>
<code>ValueError: invalid literal for int()</code>	<code>int()</code> on a decimal string
<code>NameError</code>	typo, or lowercase <code>true</code> / <code>false</code>
<code>SyntaxError</code> on a quoted line	same quote type inside and outside
braces in your output	missing the <code>f</code>

Before you push

`# name, assignment, date` · `snake_case` · no semicolons · it **runs** · calculated, not typed · **pushed**